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Page No → 5

Subject → Mathematics

Solⁿ 1 Given: $\triangle ABC \sim \triangle DCB$

$$\text{So, } \frac{AB}{DC} = \frac{BC}{CB} = \frac{AC}{BD}$$

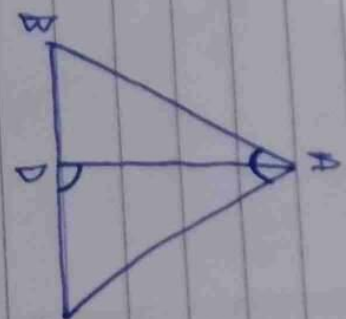
$$\Rightarrow \frac{AB}{DC} = \frac{AC}{BD} \Rightarrow AB \times BD = \underline{AC \times DC}$$

Solⁿ 2 Distance b/w origin & (a,b) is:

$$\sqrt{a^2 + b^2} = \sqrt{a^2 + b^2} \text{ units}$$

Solⁿ 3 In $\triangle ADC$ & $\triangle BAC$:

$$\angle C = \angle C \text{ (common)}$$



$$\angle ADC = \angle BAC \text{ (given)}$$

So by AA similarity, $\triangle ADC \sim \triangle BAC$

$$\text{So, } \frac{AD}{AB} = \frac{CD}{AC} = \frac{AC}{BC}$$

$$\Rightarrow \frac{CA}{CB} = \frac{AD}{AB} = \frac{CD}{AC}$$

Solⁿ 4 Section formula: $P\left(\frac{m_2x_1 + m_1x_2}{m_1 + m_2}, \frac{m_2y_1 + m_1y_2}{m_1 + m_2}\right)$

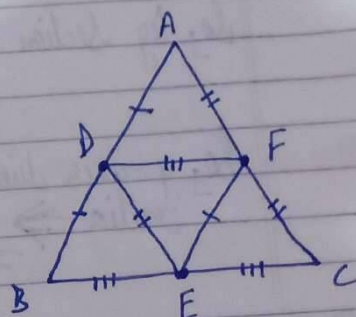
Solⁿ 5 In $\triangle EFD$ & $\triangle BED$:
 $EF = BD$ (MPT)
 $FD = BE$ (MPT)
 $DE = ED$ (common)

$\Rightarrow$ By SSS, $\triangle EFD \cong \triangle BDE$

$\rightarrow$ Similarly, $\triangle EFD \cong \triangle ADF \cong \triangle FEC$

So, areas of $\triangle EFD$, $\triangle ADF$, $\triangle FEC$ & $\triangle BDE$ are equal
 Let it be $= x \text{ cm}^2$
 So, ratio of ar. $\triangle EFD$ to ar. $\triangle ABC$ is:

$$x : 4x \Rightarrow \underline{\underline{1:4}}$$



Solⁿ 6 Let point $P(0, y)$ divide the line joining $A(-3, 2)$ and $B(6, 1)$ in the ratio $K:1$.

So, By Section Formula: $0 = \frac{-3 + 6K}{K+1} \Rightarrow 26K = 31 \Rightarrow K = \frac{31}{26}$

So, y-axis divides the line segment AB in the ratio $\Rightarrow 1:2$

Solⁿ 7 Given that $DE \parallel BC$.

So, By BPT:

$$\frac{x+4}{x+3} = \frac{2x-1}{x+1}$$

$$\Rightarrow (x+4)(x+1) = (2x-1)(x+3)$$

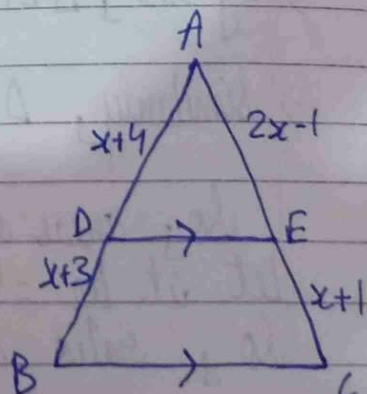
$$\Rightarrow x^2 + x + 4x + 4 = 2x^2 + 6x - x - 3$$

$$\Rightarrow x^2 + 5x + 4 = 2x^2 + 5x - 3$$

$$\Rightarrow 7 = x^2$$

$$\Rightarrow x = +\sqrt{7}, -\sqrt{7} \text{ (not possible)}$$

$\Rightarrow$ So, $x = \sqrt{7}$



2) and

Solⁿ 8 Let the point $P(x, 0)$ be equidistant from $A(-3, 4)$ and $B(2, 5)$.

$$\Rightarrow \text{So, } AP = BP$$

$$\Rightarrow \sqrt{(x+3)^2 + (-4)^2} = \sqrt{(x-2)^2 + (-5)^2}$$
$$\Rightarrow (x+3)^2 + 16 = (x-2)^2 + 25 \quad \text{\{Square both sides\}}$$

$$\Rightarrow x^2 + 9 + 6x + 16 = x^2 + 4 - 4x + 25$$

$$10x = 4$$

$$x = \frac{4}{10}$$

$$\Rightarrow \text{So, coordinates of } P = \left(\frac{2}{5}, 0 \right)$$

Solⁿ 9 By Pythagoras theorem:

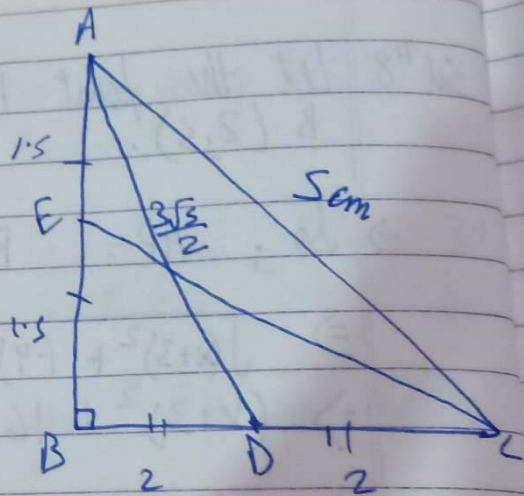
$$S^2 = (AB)^2 + (BC)^2$$

$$\Rightarrow 2S = AB^2 + BC^2$$

We know that $3, 4, 5$ is a triplet.

So, let $AB = 3$ & $BC = 4$

Now,



Solⁿ 10

Solⁿ 10 Points : $A(6, 8)$, $B(3, 7)$, $C(-2, -2)$, $D(1, -1)$

$$\rightarrow AB = \sqrt{9 + 1} = \sqrt{10}$$

$$\rightarrow BC = \sqrt{25 + 81} = \sqrt{106}$$

$$\rightarrow CD = \sqrt{9 + 1} = \sqrt{10}$$

$$\rightarrow AD = \sqrt{25 + 81} = \sqrt{106}$$

$$\rightarrow AC = \sqrt{64 + 100} = \sqrt{164} = 2\sqrt{41}$$

$$\rightarrow BD = \sqrt{4 + 64} = \sqrt{68} = 2\sqrt{17}$$

$\Rightarrow$ Therefore, opp. sides, i.e., $AB \& CD$; $AD \& BC$ are equal while the diagonals are unequal. So, the given quadrilateral is a parallelogram.